



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING(24CI3201)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Driven Emotion Detection from Text using Machine Learning	
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Abstract

The rapid growth of digital communication through social media, online reviews, messaging platforms, and customer feedback systems has generated a vast amount of textual data containing valuable emotional information. Manually analysing these emotions is time-consuming and inefficient. This project presents an automated emotion detection system that classifies textual input into emotions such as joy, sadness, anger, fear, love, and surprise using Machine Learning and Natural Language Processing (NLP) techniques. The system preprocesses text, converts it into numerical features using the TF-IDF vectorization technique, and trains a Logistic Regression model to predict emotions accurately. To introduce fundamental MLOps concepts, the project integrates MLflow for experiment tracking and model versioning, while Flask is used to deploy the trained model as a REST API. A simple web interface enables users to enter text and receive real-time emotion predictions. The project demonstrates an end-to-end workflow for building, managing, and deploying a machine learning model while maintaining simplicity suitable for academic learning.

Signature of the Guide

HOD